ANALYTICS STRATEGY / GROUP 3

Online News Popularity

Predicting sharing potential to guide article promotion

TEAM MEMBERS

Xu Yijie, Kavin Lertsattha, Yang Huichao, Zhang Zirui, Cheng Kuangmeng, Zhou Dianfeng

AN6003 Section B Group 03 - Team Project

Agenda

Business Context & Objectives [01]

Data Preparation & Exploration [02]

Logistic Regression [03]

CART [04]

Random Forest [05]

Evaluation & Recommendations [06]

Part 1

Business Context & Objectives

Presented by Xu Yijie



WE ASKED MASHABLE READERS TO PREDICT WHAT'S GOING TO HAPPEN AT THE APPLE 'SURPRISE AND SHINE' KEYNOTE EVENT.

Apple event: 77% of our readers expect iPhone price hikes

TECH

'GTA 6' Criminal Profile can reportedly break if you cause too much mayhem

MOBILE

Every rumored name for the foldable iPhone, as the 'Duo' enters the chat

MOBILE

6 products we expect at Apple's event

Many stories. Limited promotion resources.

Our ambition

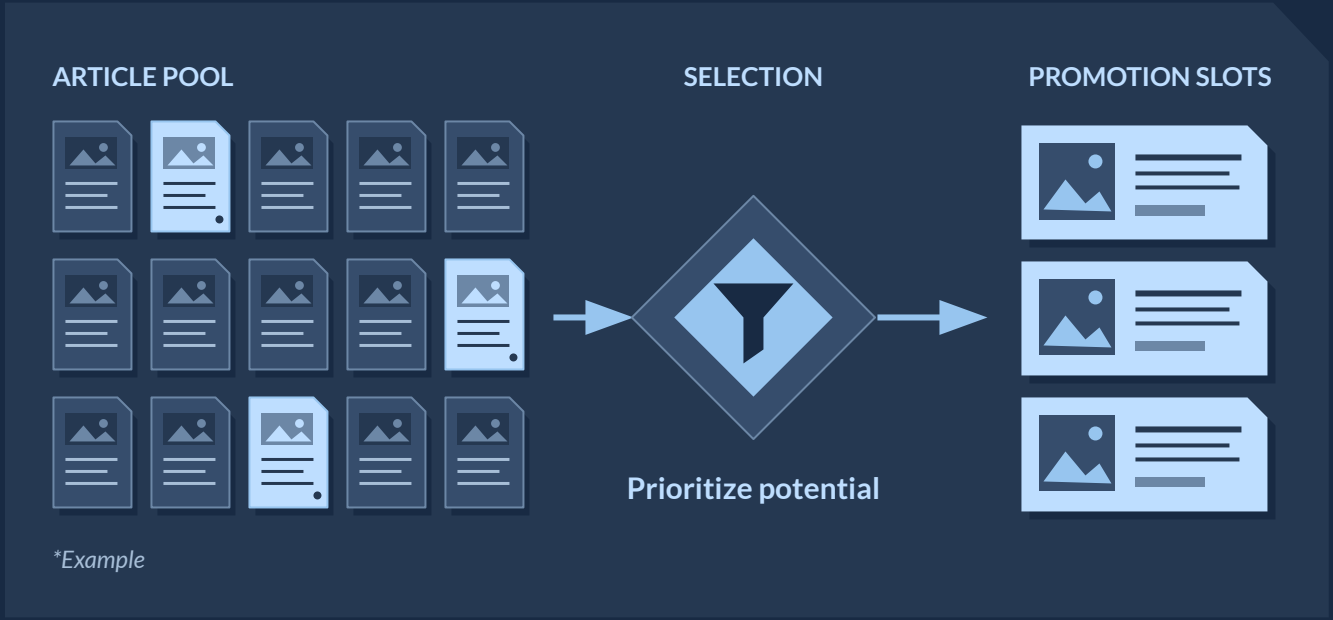
Amplify reach and create breakout stories.

Our challenge

Limited budget for digital promotion.

Our task

Identify the articles with the strongest sharing potential.



Predict Before Publication

Past articles provide the answers. New articles need a prediction.

01 LEARN FROM PAST ARTICLES

Learn how characteristics relate to sharing outcomes.

PAST ARTICLES: OUTCOMES KNOWN



Train the model
Learn from examples

39,644 original articles (2013–2014)

Use the trained model

02 SCORE BEFORE PUBLICATION

Use information available before publication.

NEW ARTICLES: SHARES UNKNOWN



Predict & Rank
By probability of being popular



Top 20%
for editorial review

popular = 1: shares > 1,400 popular = 0: shares ≤ 1,400

Source: UCI dataset. Definition follows Member 2. Top 20%: project capacity scenario.

Analysis Roadmap

We will test which approach produces the strongest shortlist.

BUILD THREE MODELS



Explore patterns

Understand the article data

- Logistic Regression
- CART
- Random Forest

Compare



Three models vs. simple baselines

Random selection

Same articles
Same 20% capacity

Recommend



An approach + a ranked

Top 20%

shortlist for editors

Our goal

Select more popular articles using the same 20% capacity.

Validation selects the approach; final testing evaluates the frozen choice.

Part 2

Data Preparation & Exploratory Analysis

Presented by Yang Huichao

DATA PREPARATION

Data Cleaning & Analytical Scope

FINAL CLEANED SAMPLE

39,643 articles retained

58 predictive variables

Original 39,644 · Removed 1 invalid lexical-ratio row

POPULAR ARTICLES

49.3%

19,561 articles

ZERO-LENGTH FLAG

1,181

2.98% · retained

Cleaning decisions

- **Remove one invalid lexical-ratio record**
  The only record outside the valid proportion range.
- **Retain zero-length content with a flag**
  Kept as a meaningful data condition, not treated as missing.
- **Consolidate unclassified articles as “Other”**
  6,134 records remain visible instead of being discarded.
- **Exclude identifiers from descriptive analysis**
  URL and timedelta support identification and ordering only.

Popular = shares > 1,400

Yang Huichao

DATA SPLIT

Three Time Blocks

Non-overlapping periods defined by timedelta ordering

TRAINING

23,793

articles · 60.0% of sample

Popular rate

50.9%

earliest 60%

VALIDATION

7,966

articles · 20.1% of sample

Popular rate

50.5%

middle 20%

TEST

7,884

articles · 19.9% of sample

Popular rate

43.5%

latest 20%

- The test-period popular rate is **7.4 percentage points** below training.
- This gap indicates **temporal distribution shift**.

Yang Huichao

PUBLISHING TIME

Weekend Uplift Appears Across Every Channel

Popular rate = popular articles ÷ all articles within each day-type group

OVERALL POPULAR RATE

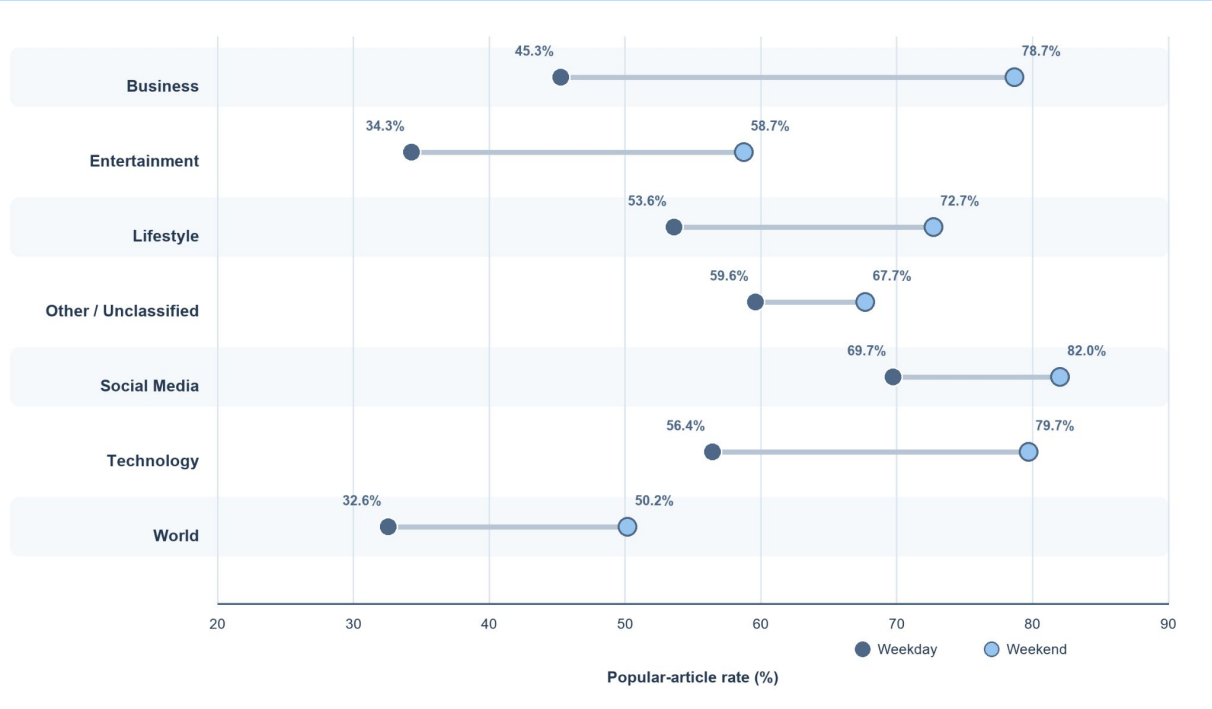
Weekday	Weekend
46.7%	67.1%

WEEKEND DIFFERENCE

+20.4 pp

- All channels show uplift.
- Largest: Business **+33.4 pp**
  Smallest: Other **+8.1 pp**
- Descriptive association only;
  it does not establish causality.

Popular rate by channel and day type



Yang Huichao

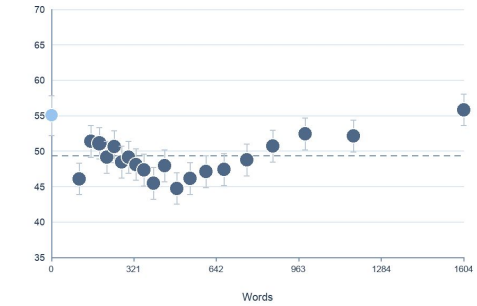
CONTENT FORMAT

Content-Format Patterns Differ by Feature

Popular rate = popular articles ÷ all articles
Point size reflects the number of observations; vertical bars show 95% confidence intervals.

Article length

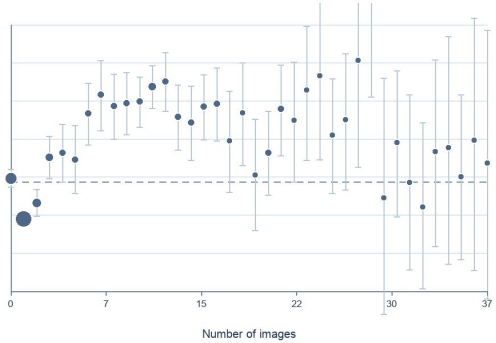
Popular rate vs. number of words



- A U-shaped pattern appears.
- Zero-length articles behave as a distinct condition.

Image count

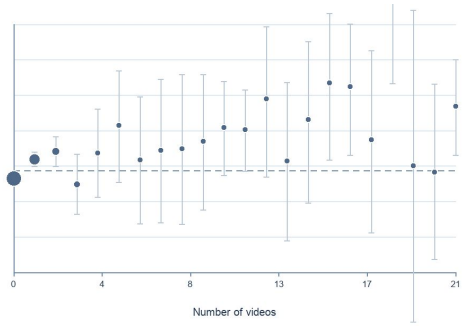
Popular rate vs. number of images



- Rates rise at low image counts.
- Variation increases in the sparse high-count tail.

Video count

Popular rate vs. number of videos



- The pattern is comparatively weak.
- Tail estimates are uncertain because observations are sparse.

Yang Huichao

Part 3

Logistic Regression

Presented by Zhou Dianfeng

LOGISTIC REGRESSION

One Model, All Relevant Factors

Why logistic regression? It estimates popularity and explains adjusted associations.

45 fields

considered together

Outcome

popular = 1 when shares > 1,400
popular = 0 otherwise

What enters the model	Fields
Sentiment and subjectivity	16
Keywords and statistics	10
Length and word use	6
Media and links	4
Topic proportions	4
Referenced-article shares	3
Channel and weekday	2

The aim: identify associations that remain after adjustment, then predict popularity.

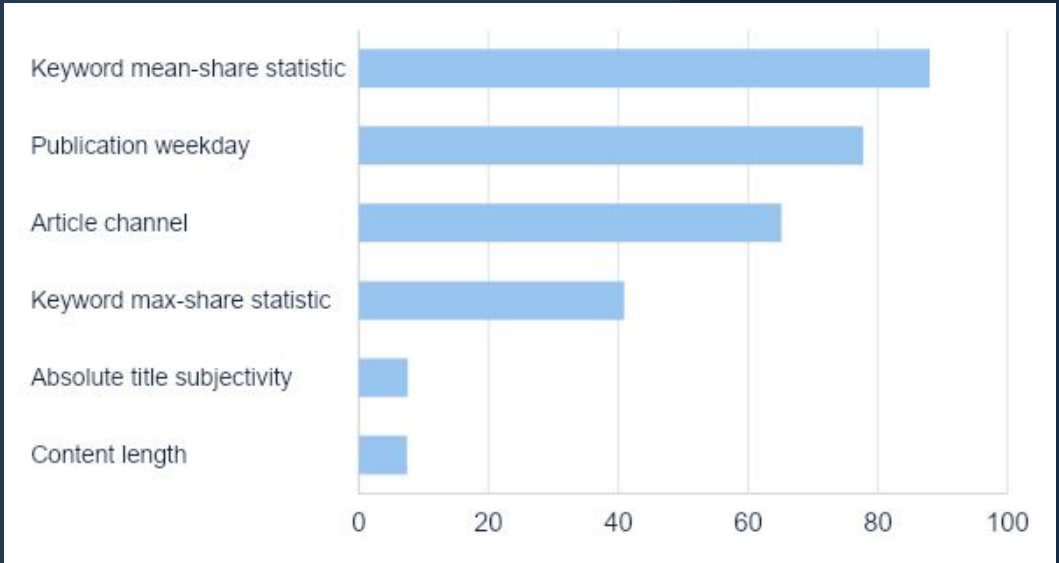
Zhou Dianfeng

LOGISTIC REGRESSION

Keywords and Publishing Context Stand Out

The six strongest signals after accounting for all other included factors

Statistical evidence: −log10(p)



Longer bars = stronger statistical evidence, not larger effects.

What this tells us

The article's context matters alongside its content.

Weekday and channel retain associations after adjustment.

All six p <0.001

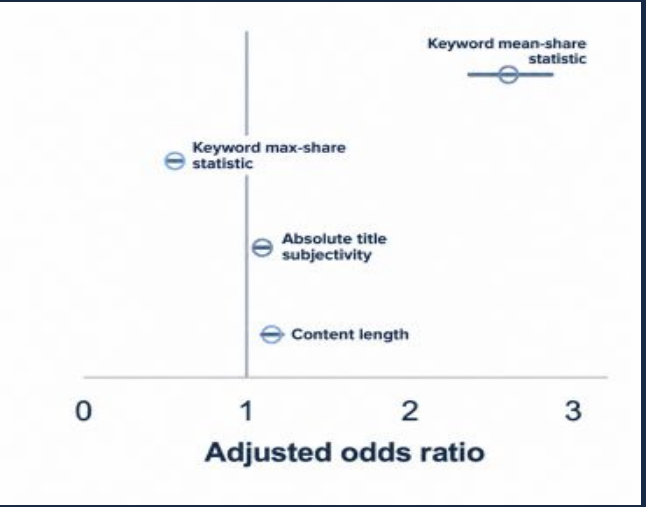
Zhou Dianfeng

LOGISTIC REGRESSION

Directions of the Leading Numeric Associations

Four numeric factors among the top six, adjusted for all other included variables

Numeric factor	OR	95% CI
Keyword mean-share statistic	2.60	2.36–2.87
Keyword max-share statistic	0.56	0.52–0.61
Absolute title subjectivity	1.10	1.06–1.13
Content length	1.15	1.10–1.22



All ORs use an increase of one training-set standard deviation.

The **subjectivity and length** effects are **smaller** positive associations.

Zhou Dianfeng

LOGISTIC REGRESSION

Classification Results at a 0.5 Cutoff

Test set confusion matrix expressed as percentages of all test articles

Confusion matrix

Actual / Predicted	Not popular	Popular	Total
Not popular	**41.12%**	15.35%	56.47%
Popular	18.54%	**24.99%**	43.53%
Total	59.67%	40.33%	100%

Shaded cells show correct classifications.

Precision asks: of the articles we call popular, how many really are?

61.95%

Precision Rate

About 62 out of every 100 articles predicted to be popular are actually popular.

Cutoff: 0.5

Probability ≥50%: predict popular.
Probability <50%: predict not popular.

Part 4

CART Model

Presented by Zhang Zirui

Knowing ≠ Knowing How to Use

Three Specific Problems We Face

? **Where's the threshold?**
How high does Metric A need to be to matter?

? **How do we judge them?**
When Metric B and C both matter, which weighs more?

? **What combination produce best prediction?**
If A+B+C = Overfitting, then choose A+B or B+C?

What We're Really Missing

A **systematic, quantifiable** method :

“knowing which metrics matter” →

“knowing how to use them”

That's why we need CART — it automatically finds the optimal thresholds and metric combinations.

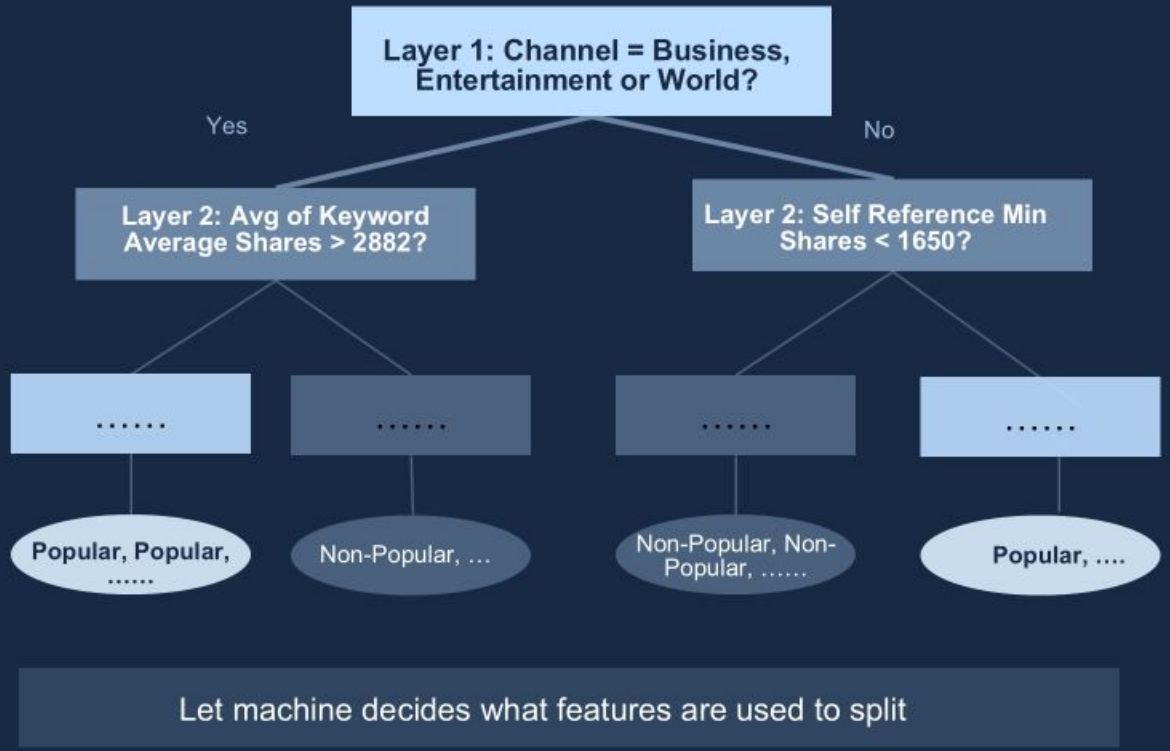
CART Logic: Like a Series of Yes/No Questions

Y Output Variable

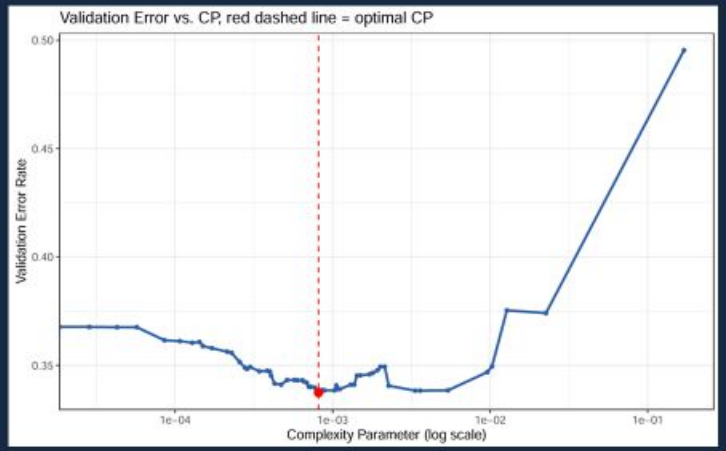
Popular: Yes or No

Xs 47 Input Variables

Features that contributes to Y



Pruning to Avoid Overfitting



When tree gets more and more complex, it start to be inaccurate

1 **The full tree overfits training noise**
It memorizes outliers and idiosyncrasies of the training data.

2 **Pruning cuts redundant branches**
Cost-complexity pruning removes splits that do not improve validation accuracy.

Number of Candidate Features

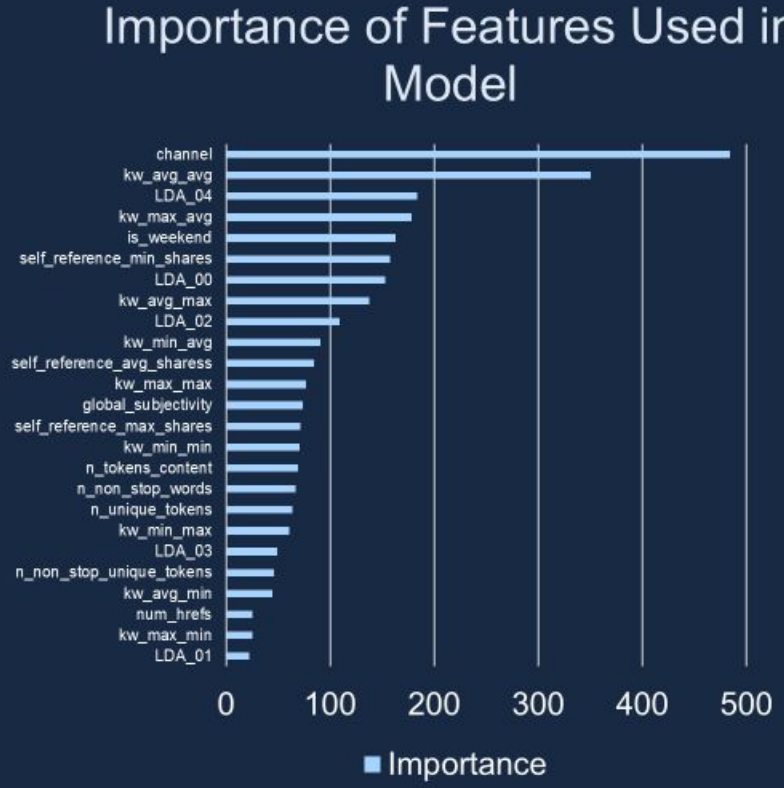
Before: 45*

After: 25

Train set to generate model
Validation set to prune
Test set to verify

*Not all 48 features in dataset are used by cart, so it's 45 not 48

Model Result: What Features Determine Popularity



Top 5 Features

The 5 strongest drivers of predicted popularity

1 **Channel**

The news channel / category the article is filed under

2 **Keyword popularity (avg)**

Mean share count of the keywords (kw_avg_avg)

3 **Topic Choosing**

The latent topic the article is grouped into (LDA_04)

4 **Keyword popularity (max)**

Highest share count among the keywords (kw_max_avg)

5 **Published on weekend**

Whether the article went out on Sat / Sun (is_weekend)

Does CART Actually Work? — vs. Guessing

Precision:

43.5% →

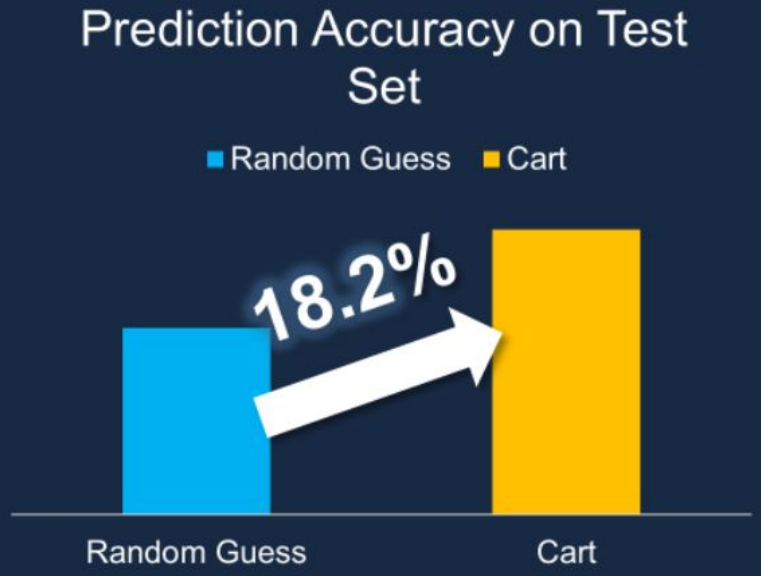
Random Guess (Baseline)

61.7%

With CART Model

Accuracy:

50.8% → 65.7%



CART isn't just theory — it does bring the improvement on predicting!

*We aim at predicting popular articles, thus here we focus precision instead of accuracy, i.e., among all articles predicted as popular, how many of them are truly popular

Part 5

Random Forest Model

Presented by Kavin Lertsattha

Presented By: Kavin Lertsattha

Random Forest Model

Can We Increase the Precision Even More?

43.5% Baseline → **61.7%** CART Model → **??.?%** RF Model

Key Variables

Output Variable

Y Popularity

Yes / No

Input Variables

X 47

Default Values

RSF = $\sqrt{47} \approx 6$

B = 500

Calibrating the Model

7,996 Records

RSF: 3 / 6 / 12

B: 100 / 300 / 500

RSF	B	OOB Error	Accuracy	False Positive Rate	False Negative Rate
3	100	36.35%	66.86%	32.77%	33.51%
3	300	34.99%	67.20%	33.27%	32.34%
3	500	34.65%	67.39%	32.77%	32.46%
6	100	35.75%	67.06%	32.11%	33.76%
6	**300**	**34.71%**	**67.90%**	**31.93%**	**32.26%**
6	500	34.51%	67.71%	32.03%	32.54%
12	100	35.96%	66.46%	32.72%	34.35%
12	300	34.70%	67.36%	32.79%	32.49%
12	500	34.48%	67.31%	33.05%	32.34%

Highest Accuracy

Lowest FPR

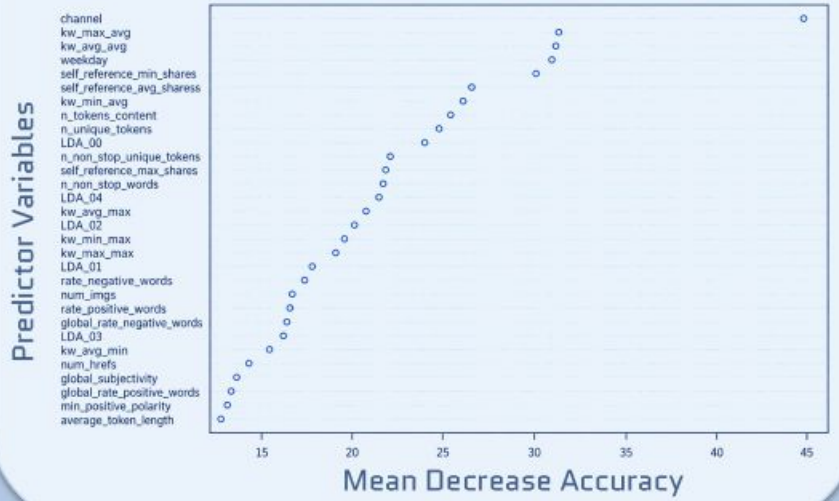
Lowest FNR

Presented By: Kavin Lertsattha

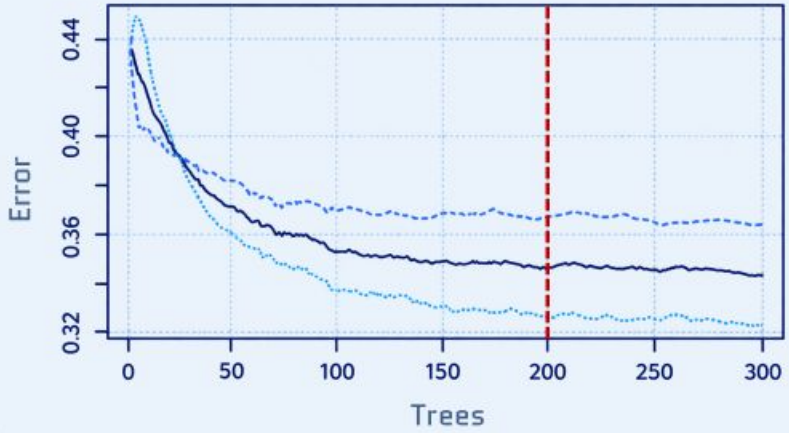
Final Model Fitting

Total Records	Number of Trees (B)	Random Subset Feature	OOB Error
31,759	300	6	33.89%

Variable Importance Plot



OOB Error Plot



OOB error stabilizes as number of trees pass 200

Top Predictors: Channel Category | Keyword Popularity | Weekday Published | Referenced Articles

Presented By: Kavin Lertsattha

Final Model Testing Results

CONFUSION MATRIX

Testing Data
7,884 Records

Actual \ Predicted	Popular	Not Popular
Popular	2,199	1,233
Not Popular	1,333	3,119

FINAL RESULTS

PRECISION	ACCURACY
62.25%	67.45%

Random Forest shown to have highest precision

Can be used as predictor for popularity and ranking

Part 6

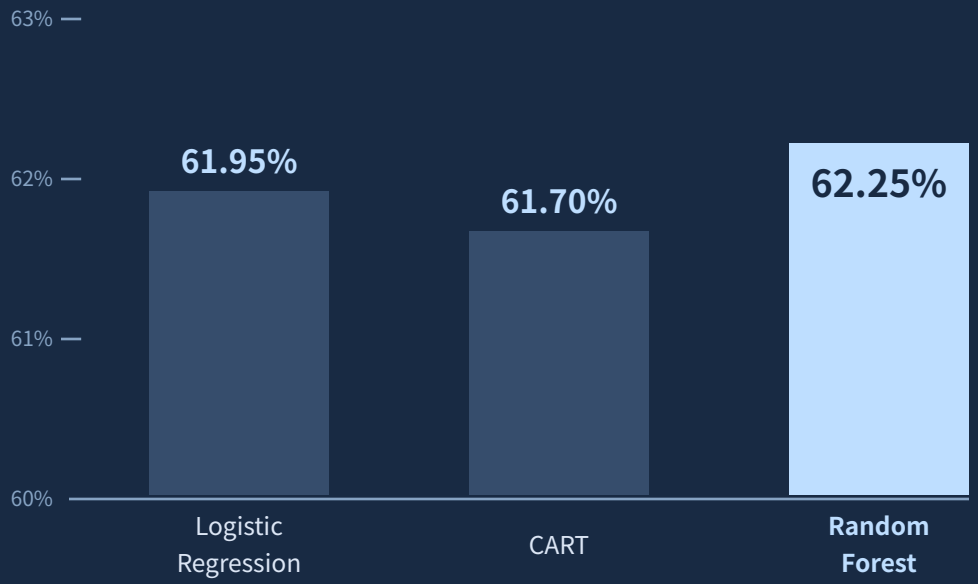
Evaluation & Recommendation

Presented by Cheng Kuangmeng

Random Forest delivers the highest precision — our recommended model

Precision = share of predicted-popular articles that are truly popular • test set, 7,884 records

PRECISION BY MODEL (%)



WHY RANDOM FOREST

62.25%

precision on held-out test set

+0.30 pp over Logistic Regression

+0.55 pp over CART decision tree

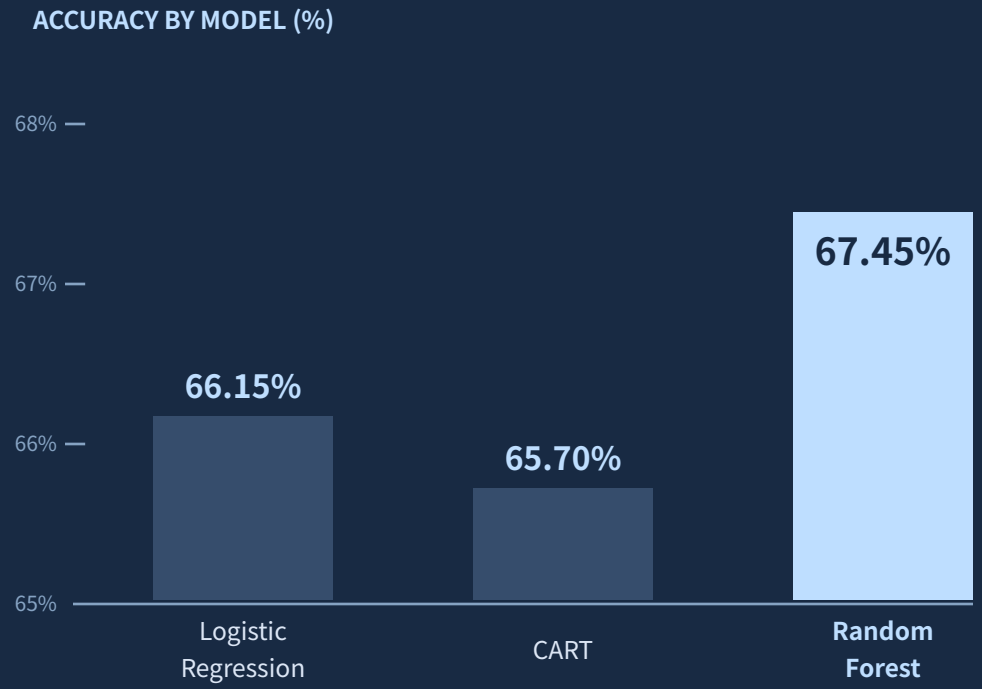
Best of the three models — chosen for deployment

Higher precision means fewer false alarms — promotion budget is spent on articles that actually go popular.

Source: Model evaluation on 7,884 held-out test records • popular = shares > 1,400 • y-axis truncated at 60%

Random Forest also leads in overall accuracy — 67.45% correct predictions

Accuracy = share of all test articles correctly classified • test set, 7,884 records



ACCURACY GAINS

67.45%

overall accuracy on test set

+1.30 pp over Logistic Regression

+1.75 pp over CART decision tree

+16.7 pp vs. 43.5% random-guess baseline

Random Forest confirms its edge on both precision and accuracy — robust for production ranking.

Source: Model evaluation on 7,884 held-out test records • popular = shares > 1,400 • y-axis truncated at 65%

Top-20% ranking hits 37.98% — 1.90x the efficiency of random selection

We score every article and promote the predicted top quintile • test set, 7,884 records

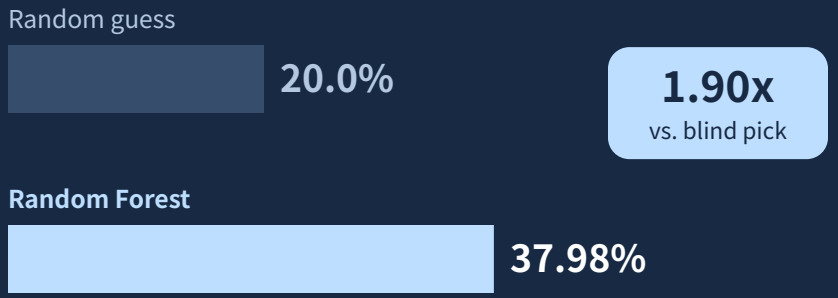
TOP-20% HIT RATE

1.90x

71.15% are popular (exceeded 1400 shares)

37.98% fell within the top 20% of actual shares, and the expected hit rate from random selection was approximately 20%

MODEL VS. BLIND SELECTION



Though 37.98% may look modest, blind selection would expect only 20% — the model nearly doubles the chance of promoting a winner.

Source: Top-quintile lift on 7,884 test records • 599 hits out of 1,577 predicted top articles • baseline = 20%

A robust, deployable model ready to guide editorial promotion decisions

Random Forest turns 58 predictive variables into a popularity score for every new article

62.25%
Precision — best of three models

67.45%
Accuracy — best of three models

1.90x
Top-20% lift vs. random selection

FOUR WAYS TO DEPLOY IT

01
Homepage slot ranking

Score every new article to prioritize which headlines earn the top editorial slots.

02
Paid promotion budget

Redirect marketing spend to model-high-score articles to maximize share-per-dollar.

03
Editorial topic guidance

Feature importance shows channel, keywords and topic drive popularity — guide pitch assignments.

04
Publishing timing

Use channel and day-type signals to recommend the optimal weekday and slot for each story.

Source: Group 3 model evaluation • 31,759 records • 300 trees • OOB error 33.89%

Thank you

Any question?

ANALYTICS STRATEGY / TEAM PROJECT

AN6003 Section B - Group 3